

Amiitesh Pradeep Kumar

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EDUCATION

Vellore Institute of Technology <i>Bachelor of Technology in Electronics and Computer Engineering</i>	Chennai, India July 2024 - July 2028
Alwin Memorial Senior Secondary School <i>12th Standard</i>	Chennai, India 2024
Alwin Memorial Senior Secondary School <i>10th Standard</i>	Chennai, India 2022

EXPERIENCE

Chairperson <i>Linux Users Group</i>	April 2026 - Present VIT Chennai, India
Technical Team Member <i>Linux Users Group</i>	2024 - 2026 VIT Chennai, India
- Provisioned VMs for the LUGVITC Cyber-0-Day event, managing distinct ISO images to support 200+ concurrent participant workstations. - Conducted a hands-on Linux technical workshop for 200+ students, instructing on core Linux fundamentals and basics bash scripting. - Made git-based CTFs for cyber-0-day	

OPEN SOURCE CONTRIBUTIONS

Contributor <i>Ly Display Manager</i>	2026
- Implemented real-time battery status display in the TUI login screen, reading from <code>/sys/class/power_supply</code>, adding a configurable <code>battery_id</code> field, localised error strings across all supported languages, and a dedicated status row merged into the v1.3.0 milestone (#826) - Fixed a UI layout bug where battery status rendered at the wrong row when both <code>hide_key_hints</code> and <code>hide_version_string</code> were enabled (#845) - Added a configurable <code>edge_margin</code> option for all screen edges, useful for curved monitor setups, resolving Zig <code>comptime_int</code> type constraints in the process (#856) - Extended power management options with a hibernate action between sleep and brightness controls, wiring it through config, lang, and the main event loop (#867)	

PROJECTS

Guzz <i>Golang, Goroutines, Concurrency</i>	Jan 2025
- Engineered a high-performance command-line file downloader in Go, leveraging Goroutines to orchestrate multi-threaded, parallel data fetching - Designed a file-chunking mechanism to divide large target files into discrete segments for concurrent downloading, safely and automatically reassembling the byte stream into the final binary to significantly accelerate download speeds	

ARM Cortex-M Bare-Metal Driver Library | *C, ARM Cortex-M3, QEMU, STM32* Aug 2025

- Implemented UART, GPIO, and SysTick timer drivers on STM32F103 with zero HAL dependencies, configuring NVIC priority groups, APB2 clock gating, and GPIO alternate-function registers via memory-mapped I/O
- Authored a custom linker script and startup assembly to zero-init BSS, copy initialised data to SRAM, and build the Cortex-M vector table before branching to main()
- Identified and eliminated an ISR ring-buffer race condition using GDB hardware watchpoints on QEMU, replacing a non-atomic pointer update with a critical-section guard

FreeRTOS Real-Time Sensor Pipeline | *C, FreeRTOS, ESP32, I2C, UART* Dec 2025

- Architected a 4-task FreeRTOS system on ESP32 (sensor acquisition, median filtering, UART logging, watchdog monitor) with queue-based IPC, achieving deterministic sub-10 ms sensor-to-log latency
- Demonstrated and resolved a priority inversion scenario using FreeRTOS priority-inheriting mutexes, documented with before/after UART timing captures as proof of correctness
- Right-sized all task stacks via uxTaskGetStackHighWaterMark() at runtime, reducing total static RAM allocation by 28%

UART Bootloader with CRC Firmware Update | *C, ARM Cortex-M, CRC-32* May 2026

- Developed a two-stage UART bootloader for STM32F103, partitioning Flash via separate linker scripts (bootloader 0x08000000, application 0x08004000) and implementing XModem-CRC firmware receive, Flash page erase, and programming in under 12 KB of bootloader code
- Implemented CRC-32 image verification before committing to Flash, with automatic rollback to the previous valid firmware on checksum failure
- Performed application handoff by relocating SCB -> VTOR, loading the application stack pointer, and branching to the reset handler — verified correct operation via GDB single-step

Power-Optimised IoT Sensor Node | *C, ESP32, FreeRTOS, MQTT, TLS, INA219* July 2026

- Engineered a battery-operated IoT node on ESP32 with ULP-assisted deep sleep and RTC wake, reducing average current from 80 mA (continuous Wi-Fi) to under 1.2 mA over a 60-second duty cycle, measured with an INA219 precision current sensor
- Implemented MQTT-over-TLS with mbedTLS session resumption, cutting per-wake TLS handshake overhead from 1.8 s to 220 ms and extending estimated 2500 mAh battery life from 3 days to 18 days

TECHNICAL SKILLS

Languages: C/C++, Golang, Zig, Python, Bash, Java

Embedded Systems: ARM Cortex-M, FreeRTOS, Zephyr RTOS, ESP32, Nordic nRF52840, QEMU

Hardware Protocols: UART, SPI, I2C, CAN bus, USB CDC, DMA, PWM/Timer, ADC

Wireless & Networking: Bluetooth LE, BLE Mesh, MQTT over TLS, POSIX Sockets, QUIC, Zigbee

System and Tools: Linux, NixOS, Git, Docker/Podman, CMake, West (Zephyr), AWS, GDB/OpenOCD, JTAG/SWD, Logic Analyzer, Wireshark